

## PRODUCT SPACES AND RANDOM PROCESSES

Let  $(\mathcal{B}, \mathcal{E})$  and  $(\mathcal{F}, \mathcal{F})$  be measurable spaces.

$$A \times B := \{(x, y) : x \in A, y \in B\}$$

product of  $A$  and  $B$

$\square$  If  $A \in \mathcal{E}$  and  $B \in \mathcal{F}$  then  $A \times B$  is called a measurable rectangle.

Then  $\mathcal{E} \otimes \mathcal{F} := \sigma(A \times B : A \times B \text{ measurable rectangle})$   
The measurable space  $(\mathcal{B} \times \mathcal{F}, \mathcal{E} \otimes \mathcal{F})$  is called the product of  $(\mathcal{B}, \mathcal{E})$  and  $(\mathcal{F}, \mathcal{F})$ .

Lemma. It is true that

$$\mathcal{E} \otimes \mathcal{F} = \sigma(\hat{\mathcal{E}} \cup \hat{\mathcal{F}})$$

where

$$\hat{\mathcal{E}} = \{A \times \mathcal{F} : A \in \mathcal{E}\}, \hat{\mathcal{F}} = \{\mathcal{B} \times B : B \in \mathcal{F}\}$$

**The infinite case:**

Let  $T$  be an arbitrary set (playing the role of index set). For each  $t \in T$ , let  $(\mathcal{B}_t, \mathcal{E}_t)$  be a measurable space. Let  $x_t \in \mathcal{B}_t$ , for each  $t \in T$ . Then  $(x_t)_{t \in T}$  (the collection) can be viewed as a function on  $T$ , especially if  $(\mathcal{B}_t, \mathcal{E}_t) = (\mathcal{B}, \mathcal{E})$  for all  $t \in T$ , because  $t \mapsto x_t$  can be seen as

a mapping

$$T \ni t \mapsto x_t \in E$$

The set  $F$  of all such functions,  $x = (x_t)_{t \in T}$  is called the product space defined by  $\{E_t, t \in T\}$

The notation, often, is

$$F = \prod_{t \in T} E_t$$

This a finite number of "positions"

A rectangle in  $F$  is a subset

$$\prod_{t \in T} A_t = \{x \in F : x_t \in A_t \text{ for each } t \in T\}$$

where  $A_t$  differs from  $E_t$  only for a finite number of  $t$ . The rectangle is measurable if

$A_t \in \mathcal{E}_t$  for every  $t$  (for which  $A_t$  differs from  $E_t$ ).

Then  $\sigma$  (measurable rectangles) is called the product  $\sigma$ -Algebra and is denoted by  $\bigotimes_{t \in T} \mathcal{E}_t$ .

The resulting product space is

$$\left( \prod_{t \in T} E_t, \bigotimes_{t \in T} \mathcal{E}_t \right).$$

In the case  $(E_t, \mathcal{E}_t) = (E, \mathcal{E})$  for each  $t \in T$  we use the notations

$$(E, \mathcal{E})^T = (E^T, \mathcal{E}^T).$$

Proposition. Let  $(\Omega, \mathcal{A})$  be a measurable space.

Let  $(F, \mathcal{F}) = \bigotimes_{t \in T} (E_t, \mathcal{E}_t)$ . For each  $t \in T$ , let  $f_t$  be a mapping from  $\Omega$  to  $E_t$ . For each  $\omega \in \Omega$ , define  $f(\omega)$  to be the point  $(f_t(\omega))_{t \in T}$  in  $F$ .

Then, the mapping  $f: \Omega \mapsto F$  is measurable relative to  $\mathcal{A}$  and  $F$  iff  $f_t$  is measurable relative to  $\mathcal{A}$  and  $\mathcal{E}_t$  for every  $t \in T$ .

This justifies the definition:

Definition. Let  $(\Omega, \mathcal{A}, P)$  be a probability space. A  $d$ -dimensional stochastic process indexed by  $I \subset [0, \infty)$  is a family of r.v.'s  $X_t: \Omega \mapsto \mathbb{R}^d$ ,  $t \in I$ . We write  $X = (X_t)_{t \in I}$ ,  $I$  is called the index set,  $\mathbb{R}^d$  the state space.

The only refinement of previous definition is that the functions  $X_t$ ,  $t \in I$  are measurable relatively to  $\mathcal{A}$  and  $\mathcal{B}(\mathbb{R}^d)$ . An object so defined is too general to be useful, especially in applications. We need more information on

$$(t, \omega) \mapsto X_t(\omega)$$

or a function of two variables. This is the reason why in the theory of probability particular classes of random processes are considered, e.g.,

- Diffusion processes
- Gaussian processes
- Markov processes
- Lévy processes
- Feller processes
- Martingales
- Semi-Markov processes
- Point processes
- . . . . .

Brownian motion  
belong to all these  
categories.

## What is Brownian motion?

If you observe plant pollen in a drop of water through a microscope, you will see an incessant, irregular movement of the particles. This was observed by Gleichen-Russumm to which the botanist Robert Brown refers when he wrote the first scientific paper on this. Brown pointed out the main features of the erratic motion of particles, qualitatively.

Brownian motion emerged in Physics, especially with Einstein. In mathematics we have contributions from

- Bochner (1908), not fully rigorous
- Wiener (1923)
- Kolmogorov
- Lévy
- Doob
- ...

We begin with a heuristic mathematical model for the erratic motion of particles (let us stay in 1-dim).

Assume that each particle

- starts at the origin
- changes position only at discrete times  $k\Delta t$  where  $\Delta t > 0$  is fixed and  $k=1, 2, \dots$
- moves  $\Delta x$  units to the left or to the right with the same probability



letting  $\Delta t$  and  $\Delta x$  to go in the appropriate way should give a random motion on the line  $\mathbb{R}$  which is continuous

let us denote by  $X_t$  the random position of a particle at time  $t \in [0, T]$ . During the time  $[0, T]$  the particle has changed its position  $N := \lfloor T/\Delta t \rfloor$  times. The decision to move left or right can be modelled by a sequence of Bernoulli r.v.'s  $\varepsilon_k, k=1, 2, \dots$  with

$$P(\varepsilon_k = 1) = 1/2 = P(\varepsilon_k = 0).$$

Hence, define

$S_N = \varepsilon_1 + \dots + \varepsilon_N \rightarrow$  number of right moves,  
 $N - S_N \rightarrow$  number of left moves,  
 so that

$$\begin{aligned} X_T &= S_N \Delta x - (N - S_N) \Delta x \\ &= 2 S_N \Delta x - N \Delta x \\ &= \sum_{k=1}^N (2 \varepsilon_k - 1) \Delta x \end{aligned} \quad (*)$$

is the position at time  $T = N \Delta t$ . Instead, for any couple of times  $t = n \Delta t$   $T = N \Delta t$ , we have

$$X_T = X_T - X_t + X_t - \underbrace{X_0}_{=0} =$$

$$= \sum_{n=1}^N (2\varepsilon_{n-1}) \Delta n + \sum_{n=1}^M (2\varepsilon_n - 1) \Delta n + \sum_{n=1}^{\infty} (2\varepsilon_{n-1}) \Delta n$$

$$= \underbrace{\sum_{n=1}^N (2\varepsilon_{n-1}) \Delta n}_{X_T - X_t} + \underbrace{\sum_{n=1}^M (2\varepsilon_n - 1) \Delta n}_{X_t - X_0}$$

Since the  $\varepsilon_n$  are iid the two increments

$X_T - X_t$  and  $X_t - X_0$  are indep. and further

$$X_T - X_t \sim X_{T-t} - X_0 \quad \text{since}$$

$$X_{T-t} = \sum_{n=1}^{N-t} (2\varepsilon_{n-1}) \Delta n \sim \sum_{n=t+1}^N (2\varepsilon_{n-1}) \Delta n$$

↓      ✓  
sum of the same number of iid r.v.'s.

Denote  $\sigma^2(t) := \text{Var } X_t$ . We have

$$\begin{aligned} \sigma^2(T) &= \text{Var}(X_T - X_t) + \text{Var}(X_t - X_0) \quad \text{indep.} \\ &= \text{Var}(X_{T-t} - X_0) + \text{Var}(X_t) \quad \text{id. distr.} \\ &= \text{Var}(X_{T-t}) + \text{Var } X_t \quad X_0 = 0 \\ &= \sigma^2(T-t) + \sigma^2(t) \end{aligned}$$

so that  $\sigma^2(t)$  is linear, i.e.,  
$$\sigma^2(T) = \sigma^2 T$$

where  $\sigma > 0$  is called the diffusion coefficient.

However we can also directly compute

$$\begin{aligned}
 \text{Var } X_T &= \text{Var} \sum_{k=1}^N (2\varepsilon_{k-1}) \Delta u \\
 &\downarrow \\
 &= \sigma^2 T \\
 &= \sum_{k=1}^N \text{Var} (2\varepsilon_{k-1}) \Delta u \quad \text{indep. of } \varepsilon_k \\
 &= \sum_{k=1}^N \text{Var} 2\varepsilon_k \Delta u \\
 &= \sum_{k=1}^N 4(\Delta u)^2 \text{Var } \varepsilon_k \\
 &= 4N (\Delta u)^2 \frac{1}{4} \quad \text{Var } \varepsilon_k = 1/4 \\
 &= \frac{T}{\Delta t} (\Delta u)^2 \quad T = N \Delta t
 \end{aligned}$$

and thus

$$\sigma^2 T = \frac{T}{\Delta t} (\Delta u)^2$$

i.e.,

$$\sigma^2 = \frac{(\Delta u)^2}{\Delta t}.$$

Using the computation above  $\circledast$  we have

$$\begin{aligned}
 X_T &= (2S_N - N) \Delta u = \frac{2S_N - N}{\sqrt{N}} \Delta u \sqrt{N} \\
 &= \frac{S_N - N/2}{\frac{1}{2}\sqrt{N}} \Delta u \sqrt{N} = \frac{S_N - \mathbb{E} S_N}{\underbrace{\sqrt{\text{Var } S_N}}_{=: S_N^*}} \Delta u \sqrt{N} \\
 &= S_N^* \sqrt{N \Delta t} \sigma \\
 &= S_N^* \sigma \sqrt{T}
 \end{aligned}$$

$$\Delta u = \sqrt{\Delta t} \sigma$$



A simple application of CLT then yields to

$$X_T = \sqrt{T} \sigma \sum_{j=1}^N \frac{N \rightarrow +\infty}{\Delta t} \sqrt{T} \sigma G$$

where  $G \sim N(0, 1)$ . Note that  $N \rightarrow +\infty$  means

$$\Delta n \rightarrow 0 \quad \Delta t \rightarrow 0, \quad \frac{\Delta x}{\sqrt{\Delta t}} = \sigma \text{ constant}$$

It follows that the particle position

$$\lim_{\Delta n, \Delta t \rightarrow 0} X_T = B_T$$

$$\text{is } N(0, T\sigma^2), \text{ while for each } t, \\ B_t \sim N(0, t\sigma^2).$$

This motivates the following.

**Definition.** A  $d$ -dimensional Brownian motion  $B = (B_t)_{t \geq 0}$  is a stochastic process indexed by  $[0, \infty)$  taking values in  $\mathbb{R}^d$  s.t.

- 0)  $B_0(\omega) = 0$  for a.e.  $\omega$
- 1)  $B_{t_n} - B_{t_{n-1}}, \dots, B_{t_1} - B_{t_0}$  are indep. for any  $n \geq 0, 0 = t_0 \leq t_1 \leq t_2 \leq \dots \leq t_n < +\infty$ .
- 2)  $B_t - B_r \sim B_{t+h} - B_{r+h}$  for all  $0 \leq r \leq t, h \geq -r$
- 3)  $B_t - B_r \sim N(0, (t-r)1)$
- 4)  $t \mapsto B_t(\omega)$  is continuous for all  $\omega$ .

We will also speak of a BM if the index set is an interval of the form  $[0, T]$ .

We say that  $(x + B_t)_{t \geq 0}$ ,  $x \in \mathbb{R}^d$ , is a  $d$ -dimensional BM started at  $x$ .

As we will see the definition is redundant, i.e.,

0) - 3)  $\rightarrow$  4) for a.a.  $\omega$

0) - 2) and 4)  $\rightarrow$  3)

Now we prove the equivalence of a different formulation of the notion "independent increments" which pass through the use of generated  $\sigma$ -algebras and will be useful later.

Then we should address the problem of finding a process fitting the definition, but we leave this for later. Before we study some properties of BM which follow from the definition and have a direct relation with the fact that BM is a Gaussian process.

Here is an equivalent formulation of the notion of "indep. increments".

Theorem. Let  $(X_t)_{t \geq 0}$  be stochastic process in  $\mathbb{R}^d$ .  
Then

$$X_t \text{ has indep. increments} \iff X_t - X_r \text{ is indep from } \sigma(X_u, u \leq r) \text{ for any } r \geq 0$$

Recall that

$$\sigma(X_u, u \leq r) := \sigma\left(\bigcup_{u \leq r} \sigma(X_u)\right)$$

Proof. We use Bochner's theorem, i.e., for  $X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$   
 $E e^{i \langle \xi, X \rangle} = E e^{i \langle \xi_1, X_1 \rangle} E e^{i \langle \xi_2, X_2 \rangle} \iff X_1, X_2 \text{ independent}$

Proof of " $\Leftarrow$ ":

$$E e^{i \sum_{k=1}^n \langle \xi_k, X_{t_k} - X_{t_{k-1}} \rangle} = E \left[ E \left[ \exp \left( i \sum_{k=1}^n \langle \xi_k, X_{t_k} - X_{t_{k-1}} \rangle \right) \middle| \mathcal{F}_{t_{n-1}}^X \right] \right]$$

$X_{t_k} - X_{t_{k-1}}$  is  $\mathcal{F}_{t_{n-1}}^X$ -measurable for  $k \leq n-1$

tower property

$\sigma(X_u, u \leq t_{n-1})$

$$= E \left[ \exp \left( i \sum_{k=1}^{n-1} \langle \xi_k, X_{t_k} - X_{t_{k-1}} \rangle \right) E \left[ \exp \left( i \langle \xi_n, X_{t_n} - X_{t_{n-1}} \rangle \right) \middle| \mathcal{F}_{t_{n-1}}^X \right] \right]$$

$$= E \left[ \exp \left( i \sum_{k=1}^{n-1} \langle \xi_k, X_{t_k} - X_{t_{k-1}} \rangle \right) \right] E \left[ \exp \left( i \langle \xi_n, X_{t_n} - X_{t_{n-1}} \rangle \right) \right]$$

$\downarrow$   
 $X_{t_n} - X_{t_{n-1}}$  indep of  $\mathcal{F}_{t_{n-1}}^X$  by hypotheses.

$$= \prod_{i=1}^m \exp(i \langle \vec{z}_k, X_{t_k} - X_{t_{k-1}} \rangle)$$

Hence the implication follows by Kac's theorem.

Proof of " $\Rightarrow$ "

$\Gamma_{NB}: \bigcup_{n \geq 0} \sigma(X_n)$  is not a  $\sigma$ -system

LEMMA: We have that

$$\mathcal{G}_\pi := \bigcup_{n \in \mathbb{N}} \bigcup_{0 \leq \alpha_1 \leq \dots \leq \alpha_n \leq \pi} \sigma(X_{\alpha_1}, \dots, X_{\alpha_n})$$

is a  $\sigma$ -system which generates  $\mathcal{F}_\pi^X \rightarrow \sigma$  (Gubenko)

Therefore it is sufficient to check independence on the  $\sigma$ -system, i.e., to check that  $\sigma(X_t - X_\pi)$  is indep from  $\mathcal{G}_\pi$  or equivalently

$X_t - X_\pi$  is indep from  $(X_{\alpha_1}, \dots, X_{\alpha_n})$ , for any  $n \in \mathbb{N}$  and  $0 \leq \alpha_1 \leq \dots \leq \alpha_n \leq \pi$ .

First we prove that  $\mathcal{G}_\pi$  is a  $\sigma$ -system.

If  $A \in \mathcal{G}_\pi$  and  $B \in \mathcal{G}_\pi$  there exist  $m, n \in \mathbb{N}$  and  $0 \leq \alpha_1 \leq \dots \leq \alpha_m \leq \pi$  and  $0 \leq \beta_1 \leq \dots \leq \beta_n \leq \pi$  such that  $A \in \sigma(X_{\alpha_1}, \dots, X_{\alpha_m})$   $B \in \sigma(X_{\beta_1}, \dots, X_{\beta_n})$ .

Let  $0 \leq u_1 \leq \dots \leq u_N \leq \pi$  be an increasing enumeration of the refinement  $\{\alpha_j, j \in m\} \cup \{\beta_j, j \in n\}$  so that

$A, B \in \sigma(X_{u_1}, \dots, X_{u_N})$  and thus

$$A \cap B \in \sigma(X_{u_1}, \dots, X_{u_N}) \subset \mathcal{G}_\pi$$

This proves that  $G_r$  is a  $\mathcal{F}_r^x$ -system. Now we prove that it generates  $\mathcal{F}_r^x$ .

It is clear that  $(x_{r_1}, \dots, x_{r_n})$  is  $\mathcal{F}_r^x$ -measurable and  $\sigma(x_{r_1}, \dots, x_{r_n}) \subset G_r$  which implies that  $\sigma(G_r) \subset \mathcal{F}_r^x$ . Furthermore, by definition of  $G_t$ , we have  $\sigma(x_t) \subset G_r$  for each  $t \in [0, r]$  so that  $x_t$  is  $G_r$ -measurable for each  $t \in [0, r]$ .

Since  $\mathcal{F}_r^x$  is the smallest  $\sigma$ -algebra which makes all the  $x_t, t \leq r$  measurable we find that  $\mathcal{F}_r^x \subset \sigma(G_r)$ .

We conclude that  $\mathcal{F}_t^x = \sigma(G_t)$ .

We are now ready to prove that  $x_t - x_r$  and  $(x_{r_1}, \dots, x_{r_n})$  are independent.

By assumption the r.v.'s  $x_{r_1} - x_{r_0}, \dots, (x_{r_n} - x_{r_{n-1}})$ ,  $x_t - x_r$  are independent. Since

$$x_{r_k} = \sum_{j=1}^{k-1} (x_{r_j} - x_{r_{j-1}}), \quad k=1, \dots, n,$$

we see that  $(x_t - x_r)$  indep of  $(x_{r_1}, \dots, x_{r_n})$ .